

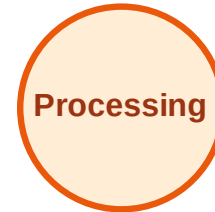
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

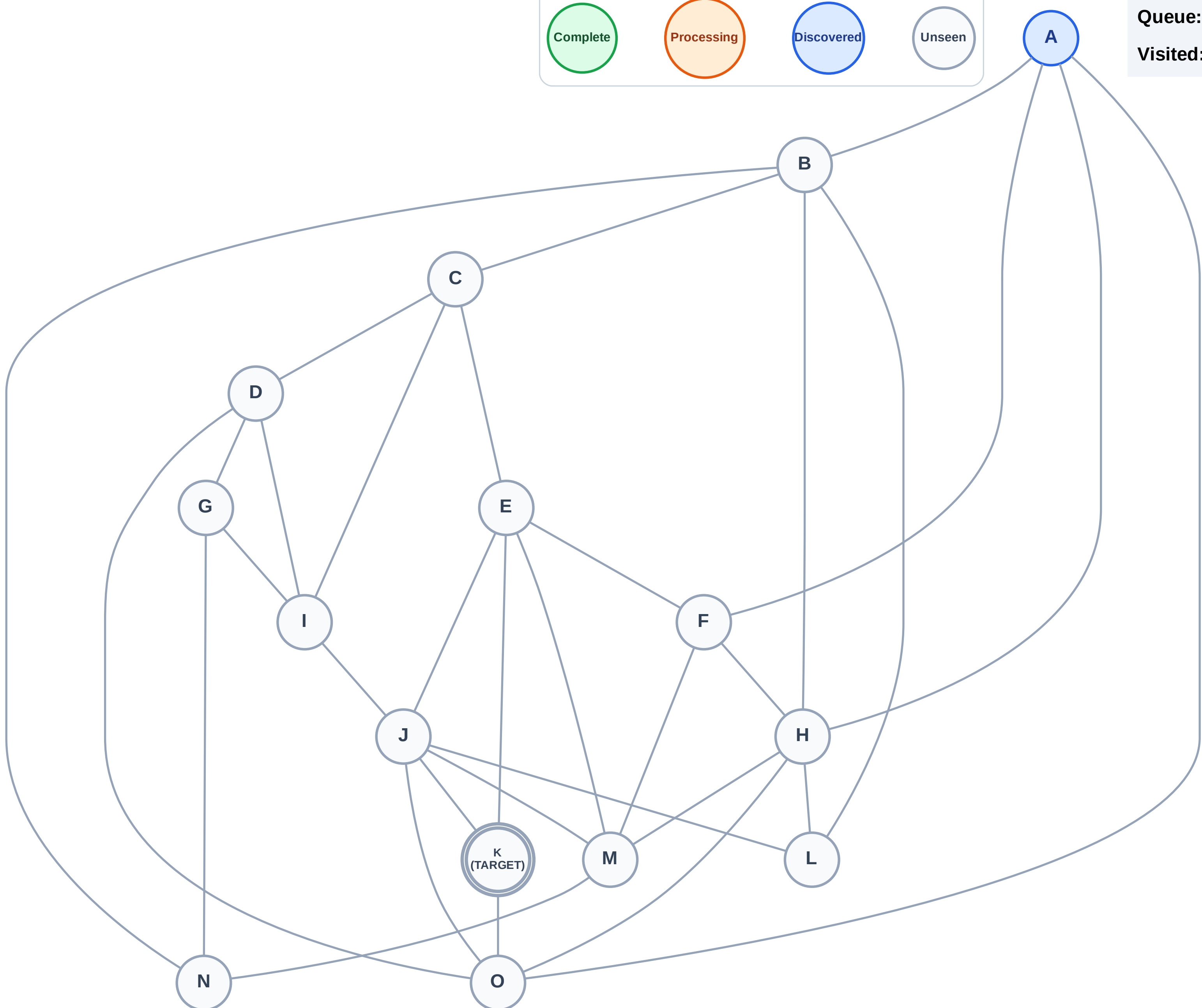


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

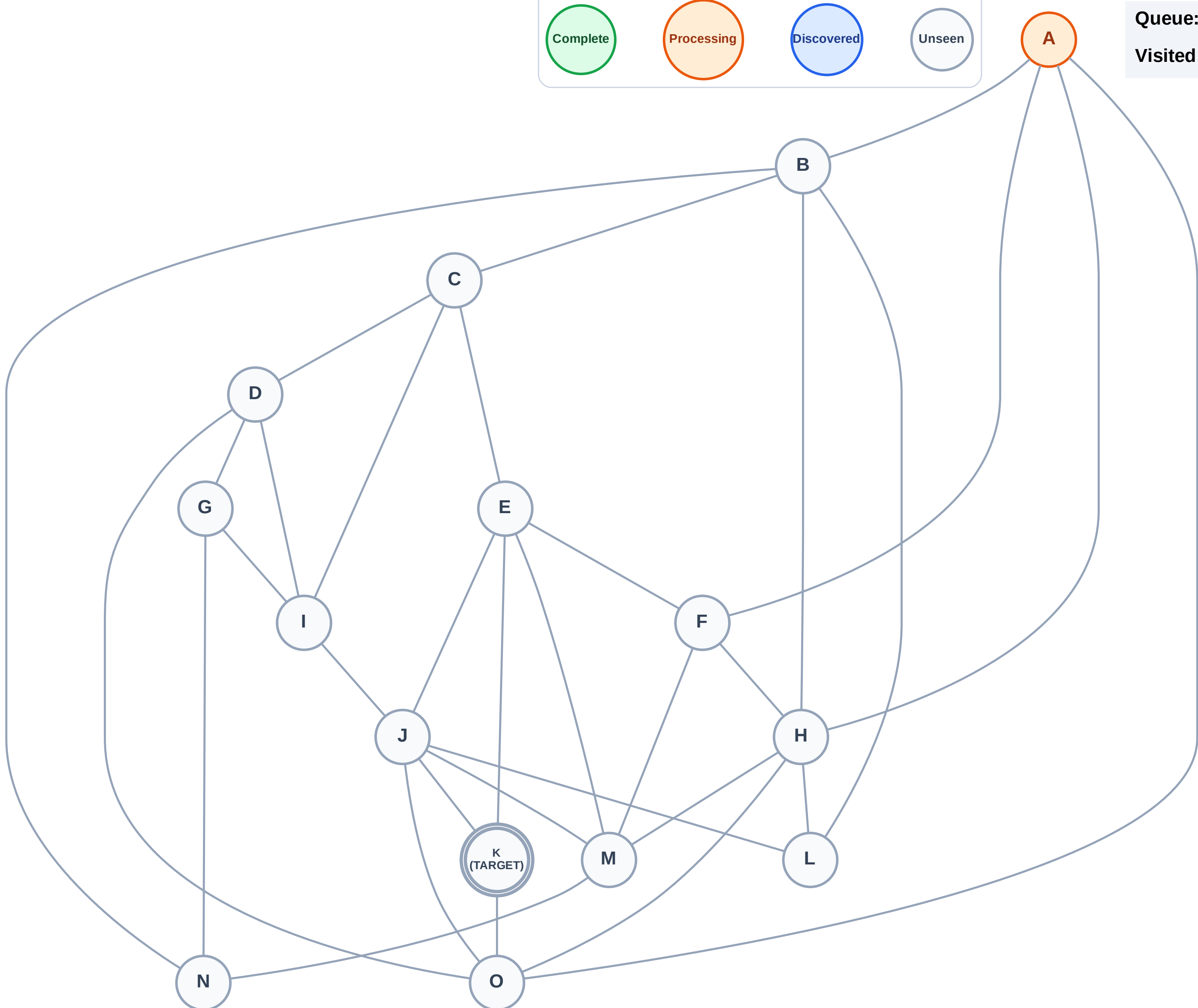
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

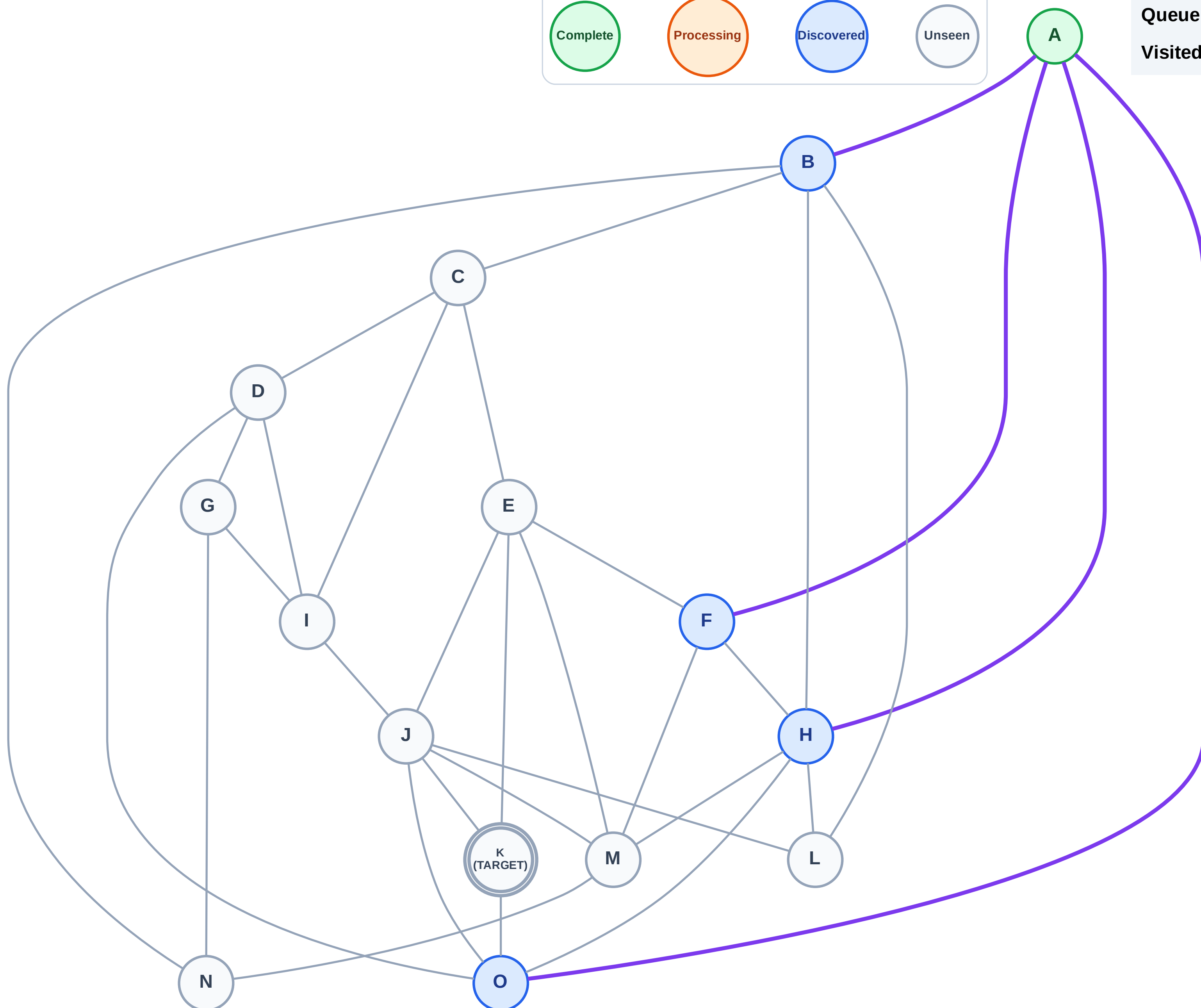
Step 3: Discover B, F, H, O from A

Legend



Queue: B → F → H → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

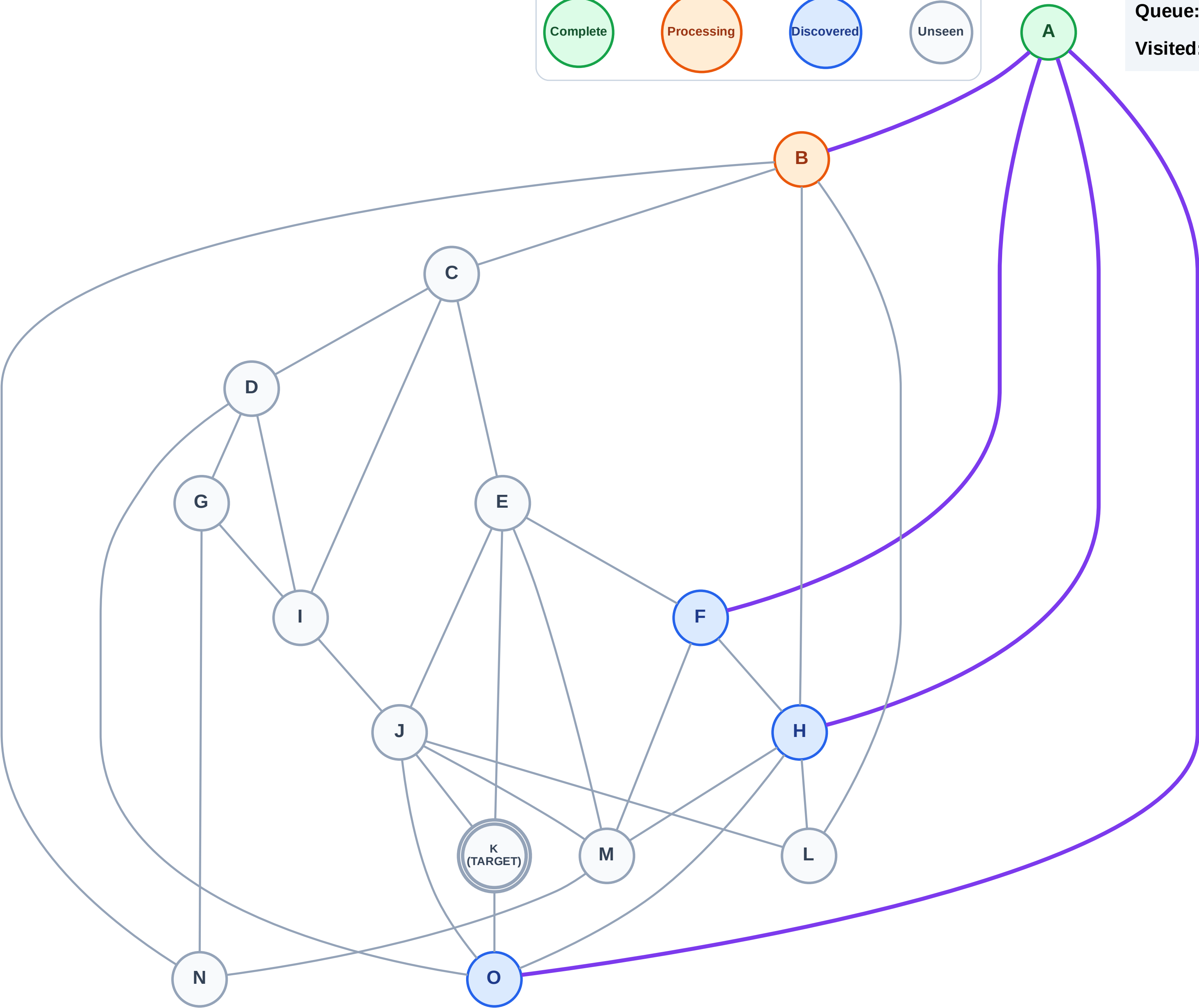
Complete

Processing

Discovered

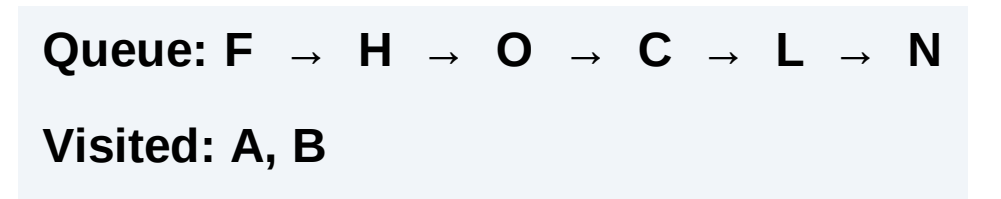
Unseen

Queue: F → H → O
Visited: A



Step 5: Discover C, L, N from B

Step 5: Discover C, L, N from B



BFS Traversal

Step 6: Process node F

Legend

Complete

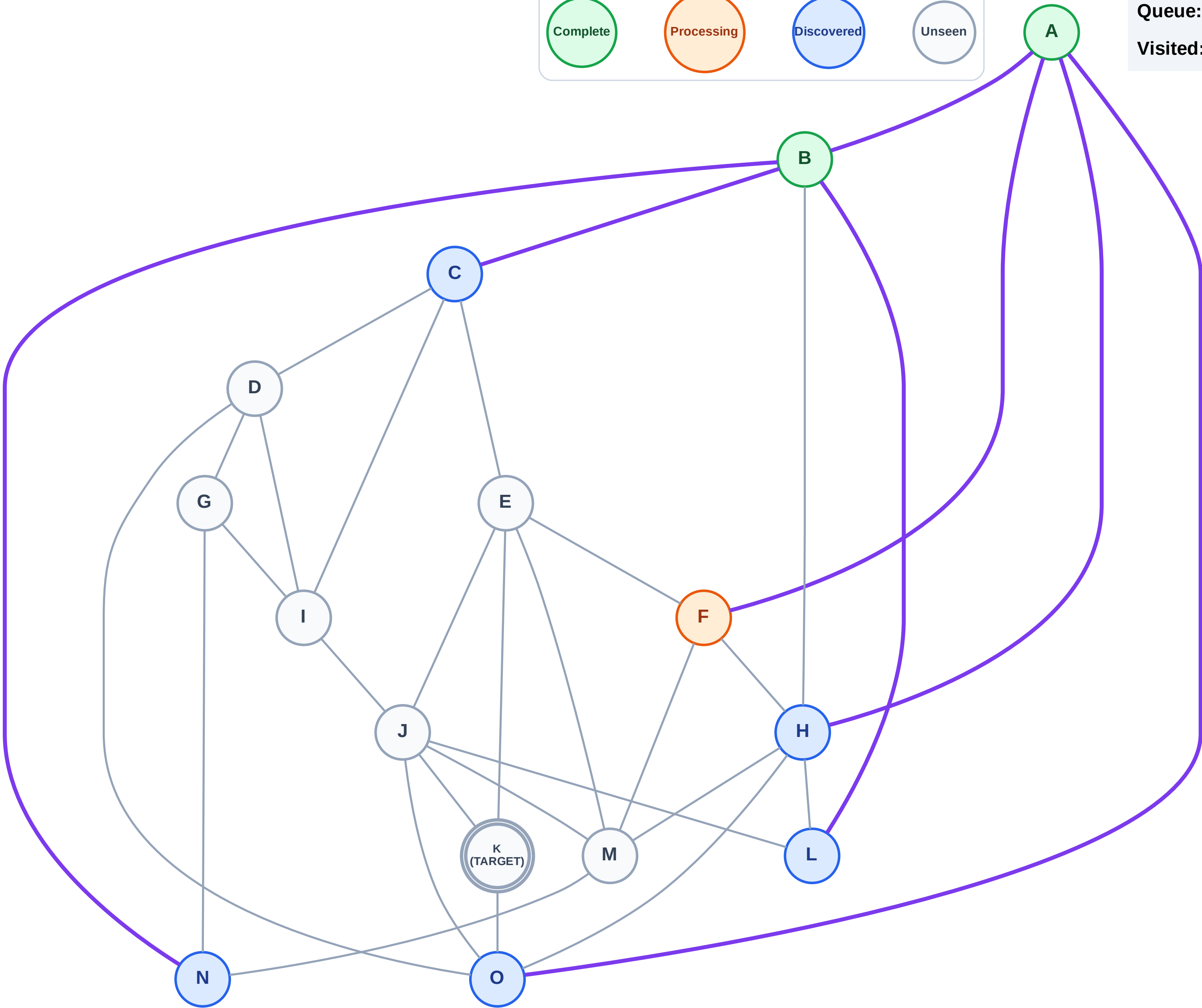
Processing

Discovered

Unseen

Queue: H → O → C → L → N

Visited: A, B



BFS Traversal

Step 7: Discover E, M from F

Legend



Complete



Processing



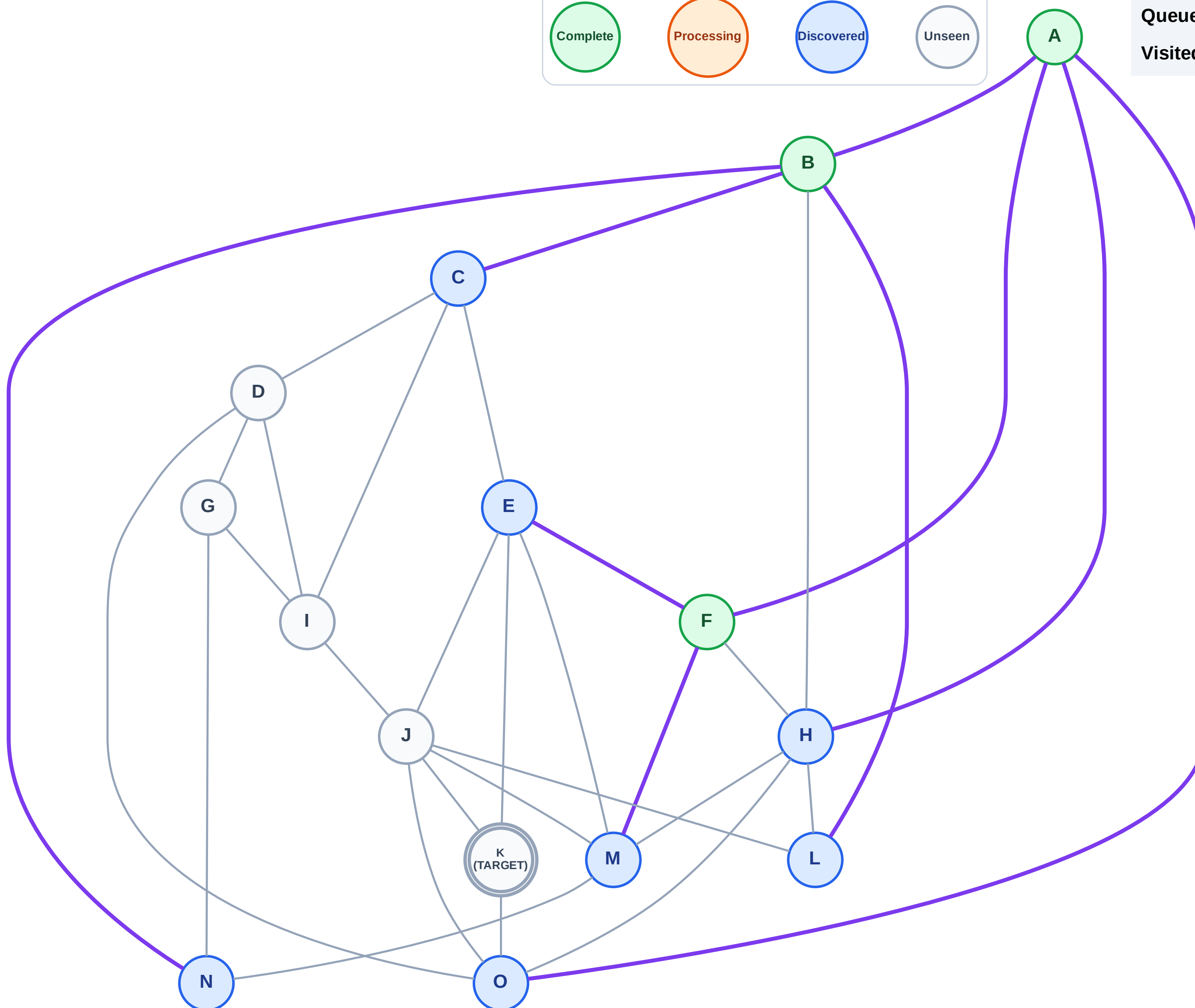
Discovered



Unseen

Queue: H → O → C → L → N → E → M

Visited: A, B, F



BFS Traversal

Step 8: Process node H

Legend

Complete

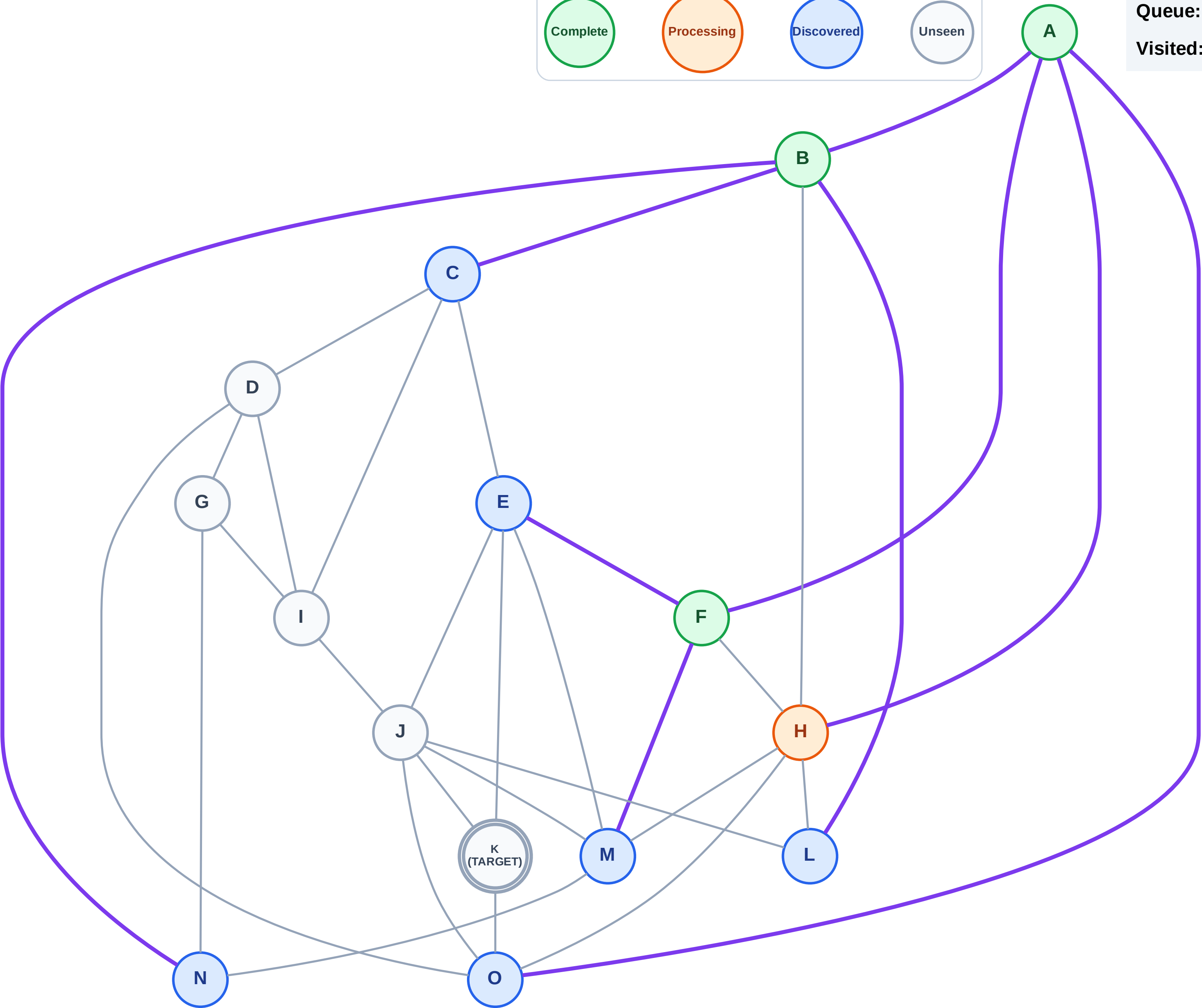
Processing

Discovered

Unseen

Queue: O → C → L → N → E → M

Visited: A, B, F



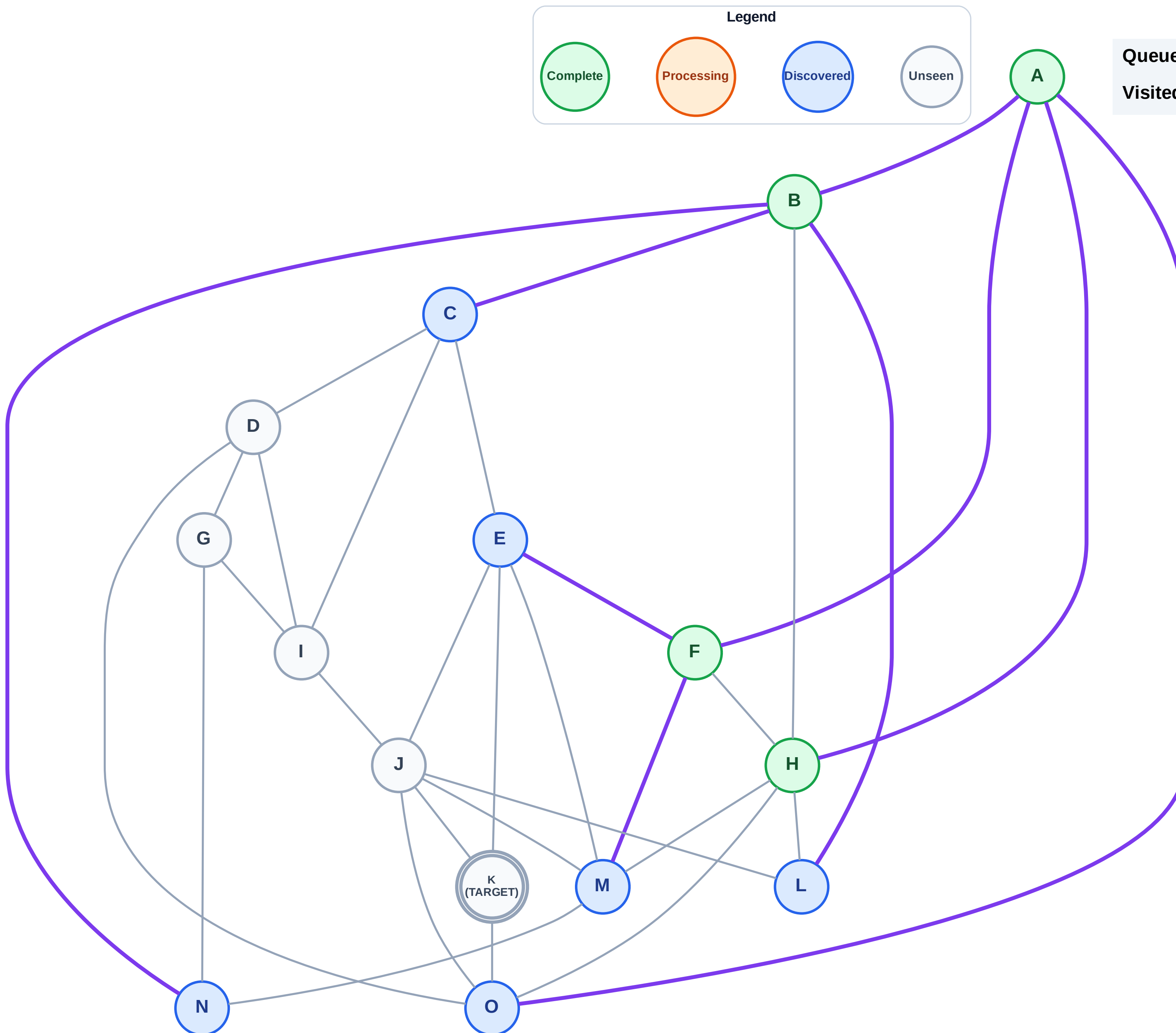
BFS Traversal

Step 9: No new neighbors from H

Legend



Queue: O → C → L → N → E → M
Visited: A, B, F, H



BFS Traversal

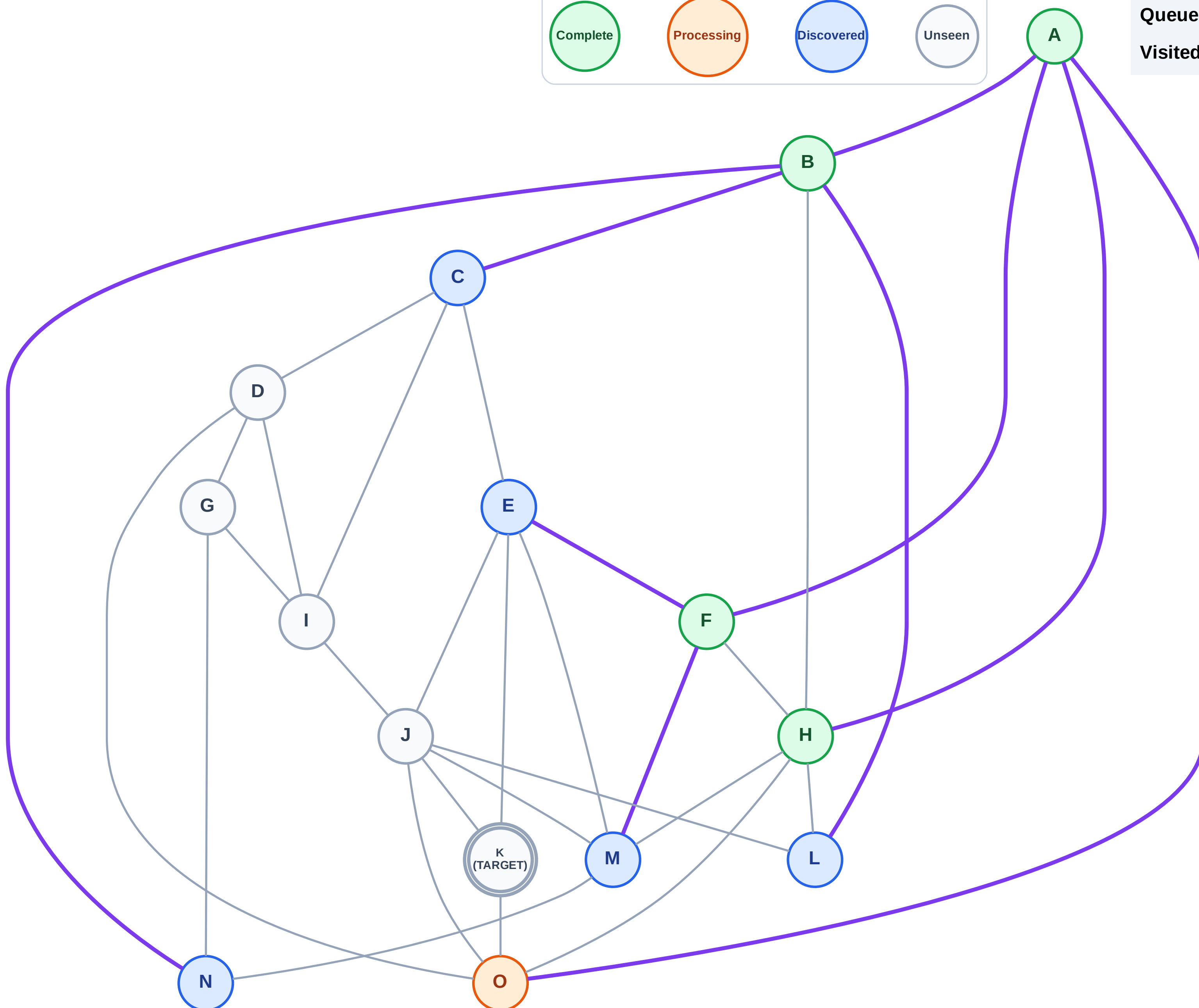
Step 10: Process node O

Legend



Queue: C → L → N → E → M

Visited: A, B, F, H



BFS Traversal

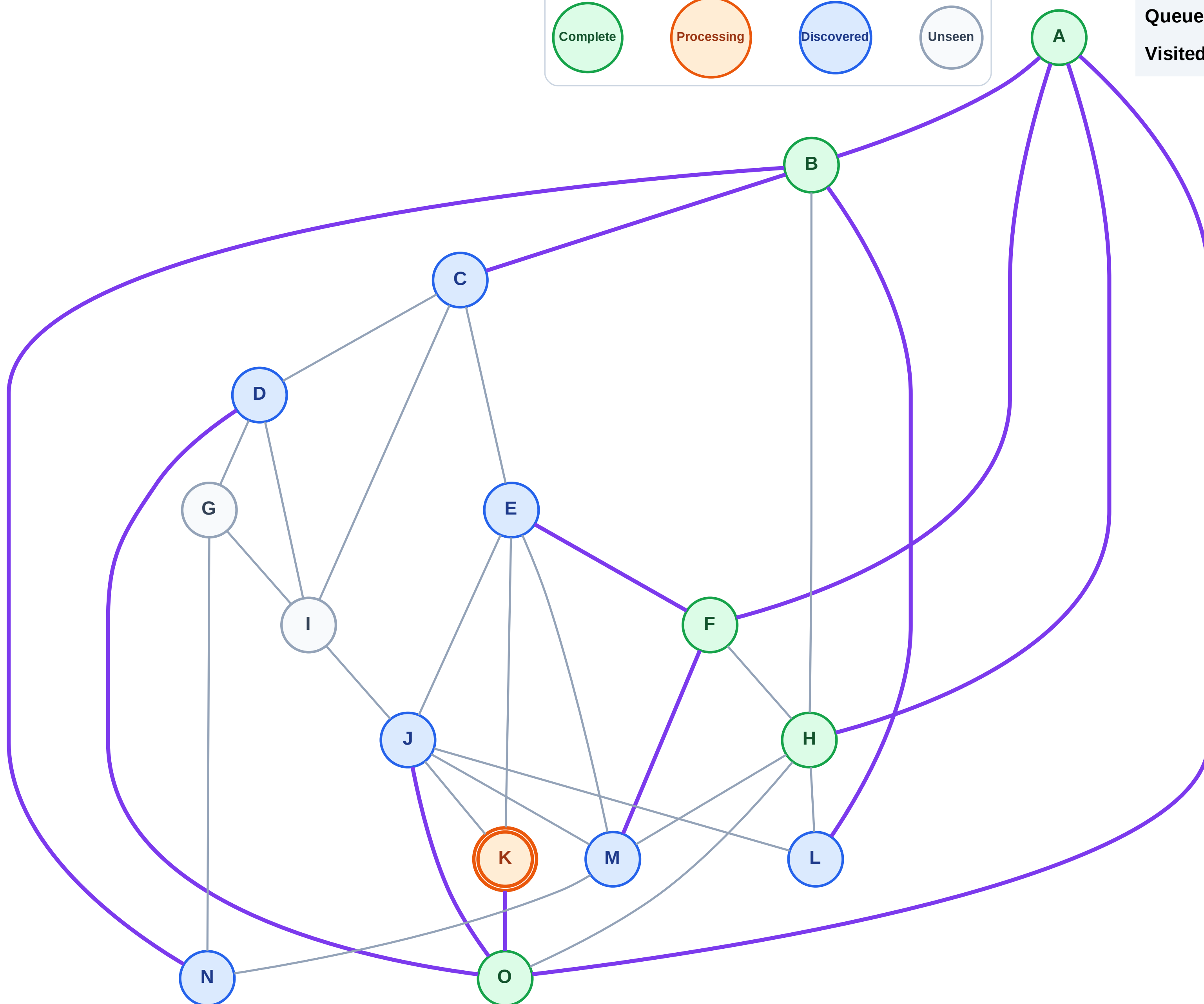
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → O → K

Legend



Queue: C → L → N → E → M → D → J

Visited: A, B, F, H, K, O



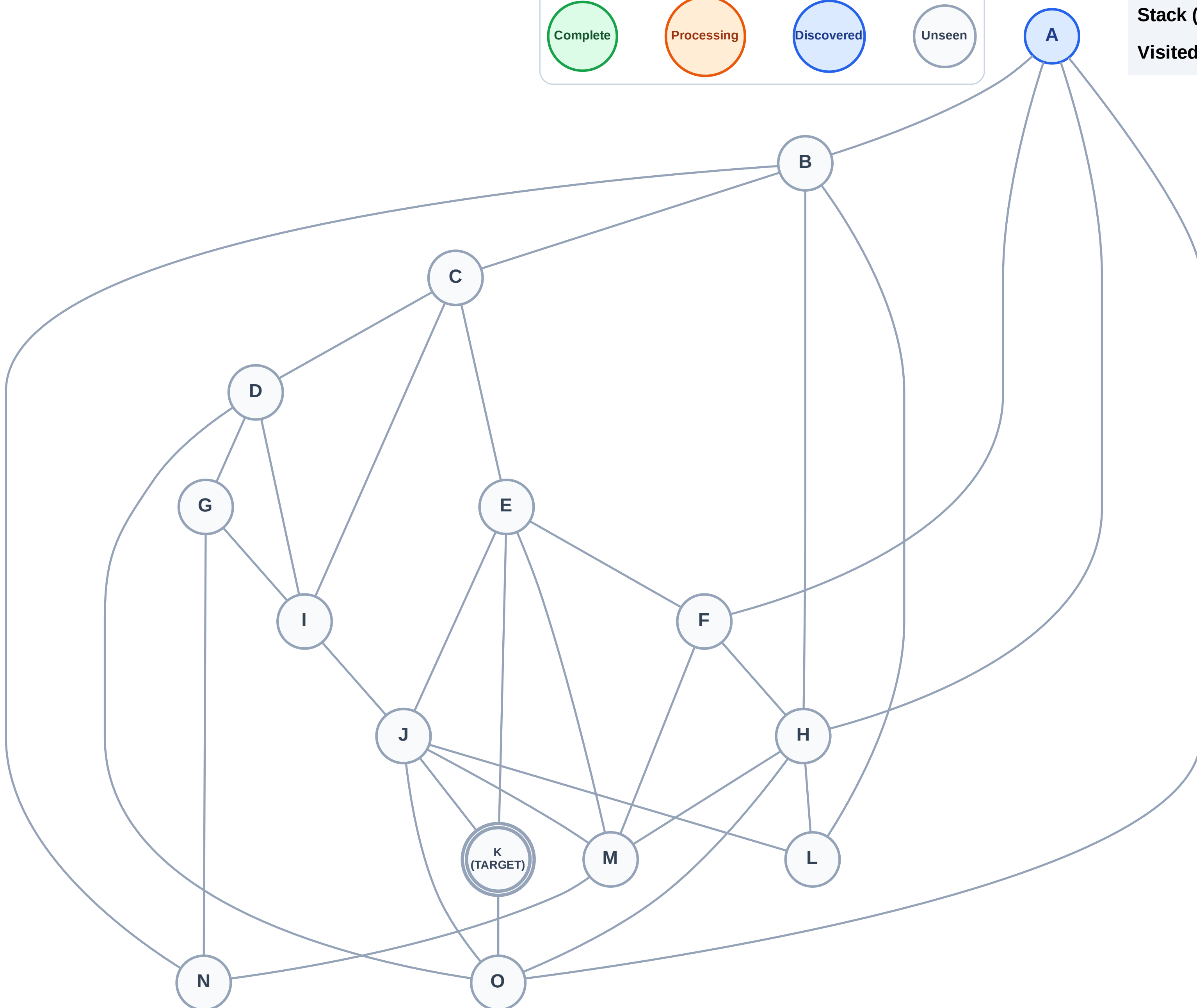
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

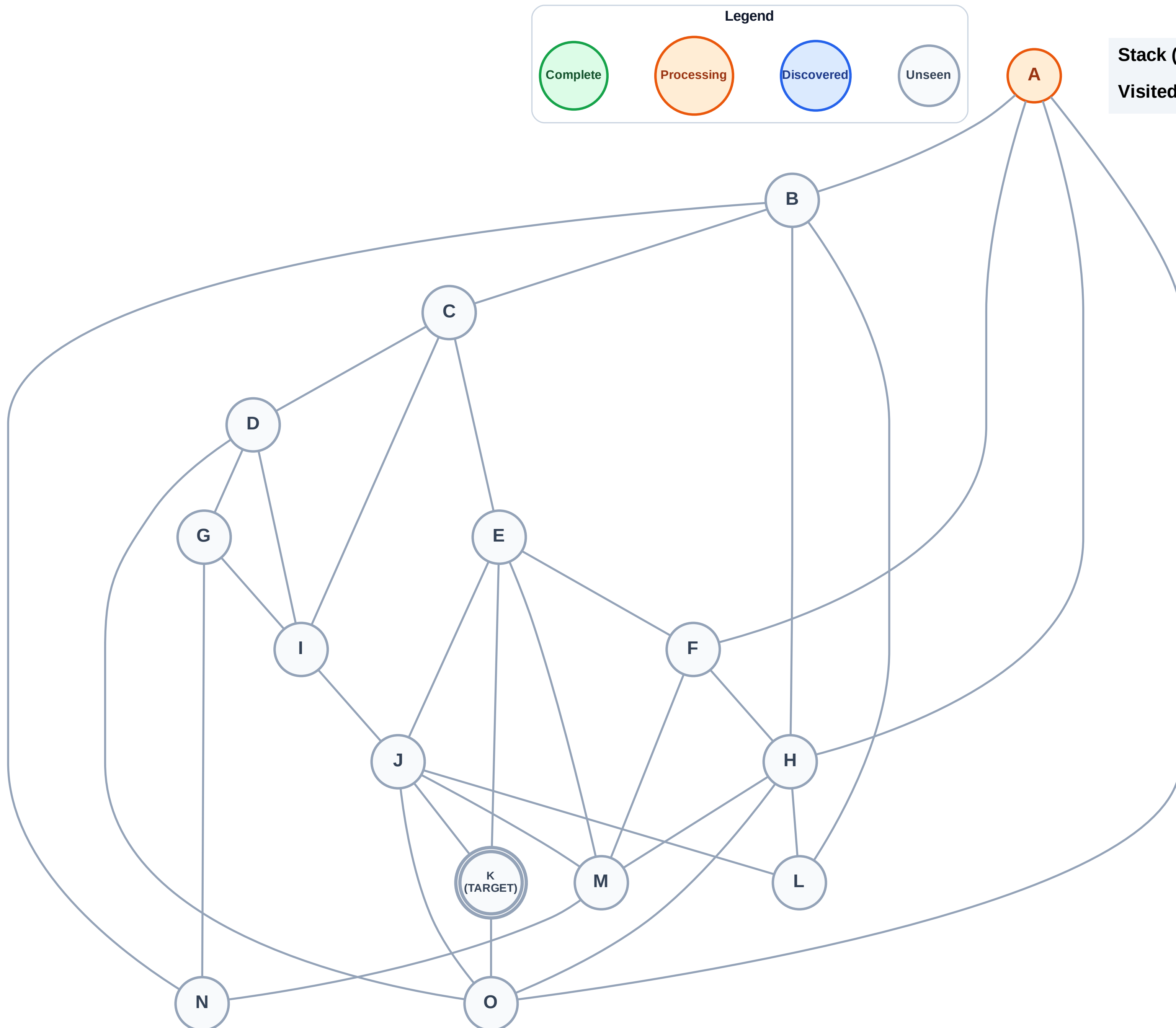
Step 2: Process node A

Legend



Stack (top at right): (empty)

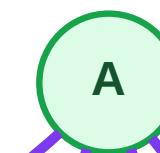
Visited: (none)



DFS Traversal

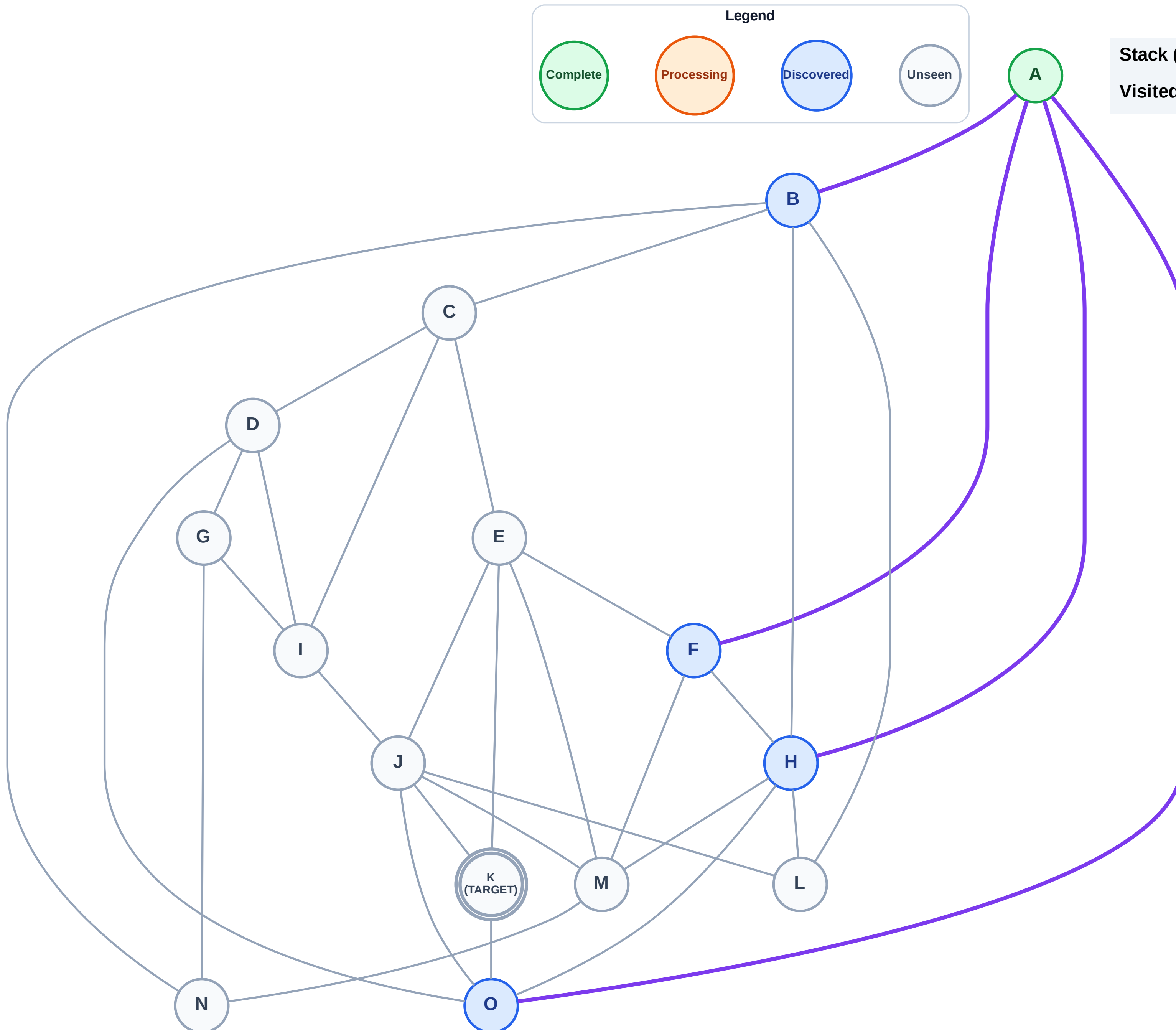
Step 3: Discover O, H, F, B from A

Legend



Stack (top at right): O | H | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

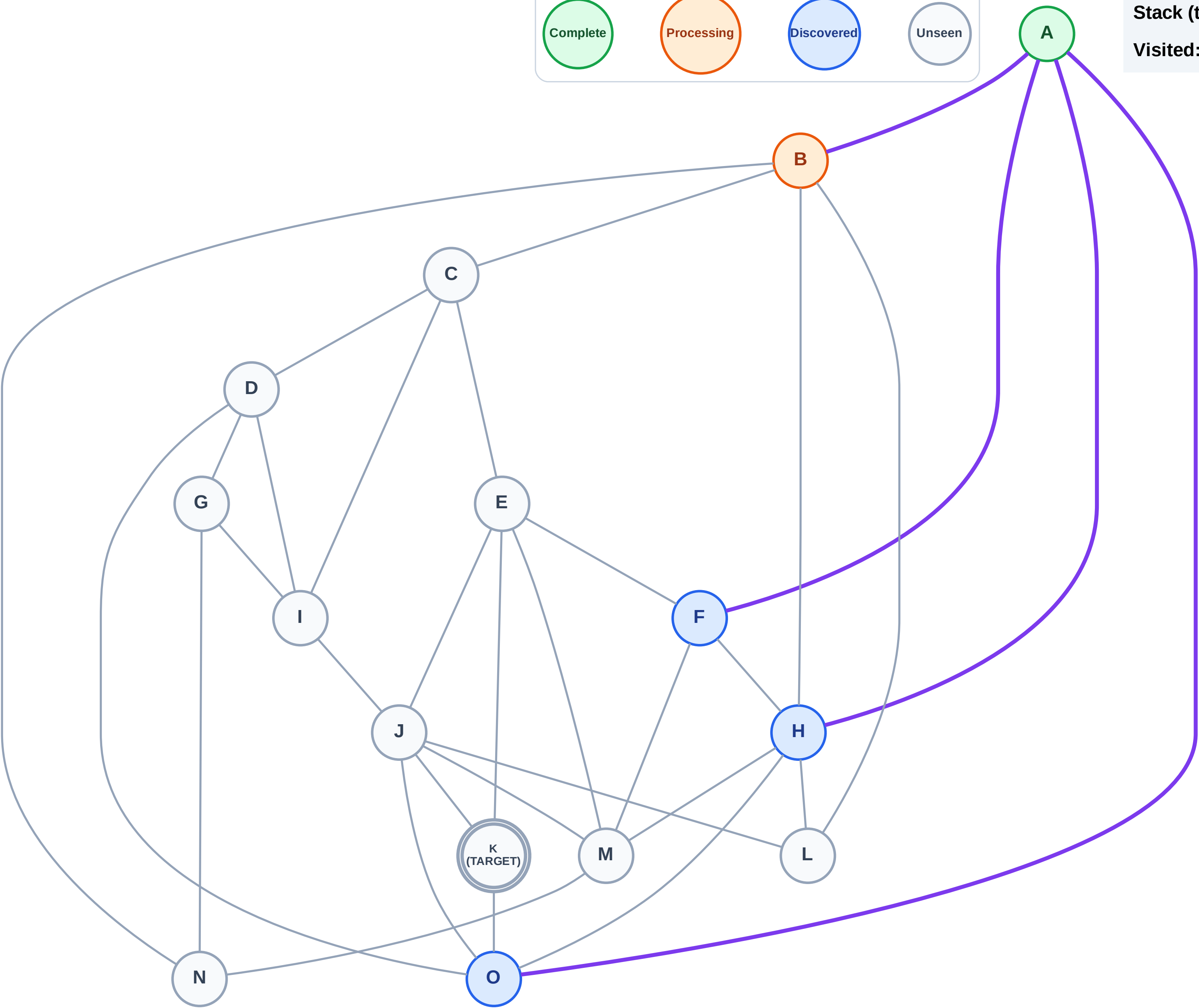
Processing

Discovered

Unseen

Stack (top at right): O | H | F

Visited: A



DFS Traversal

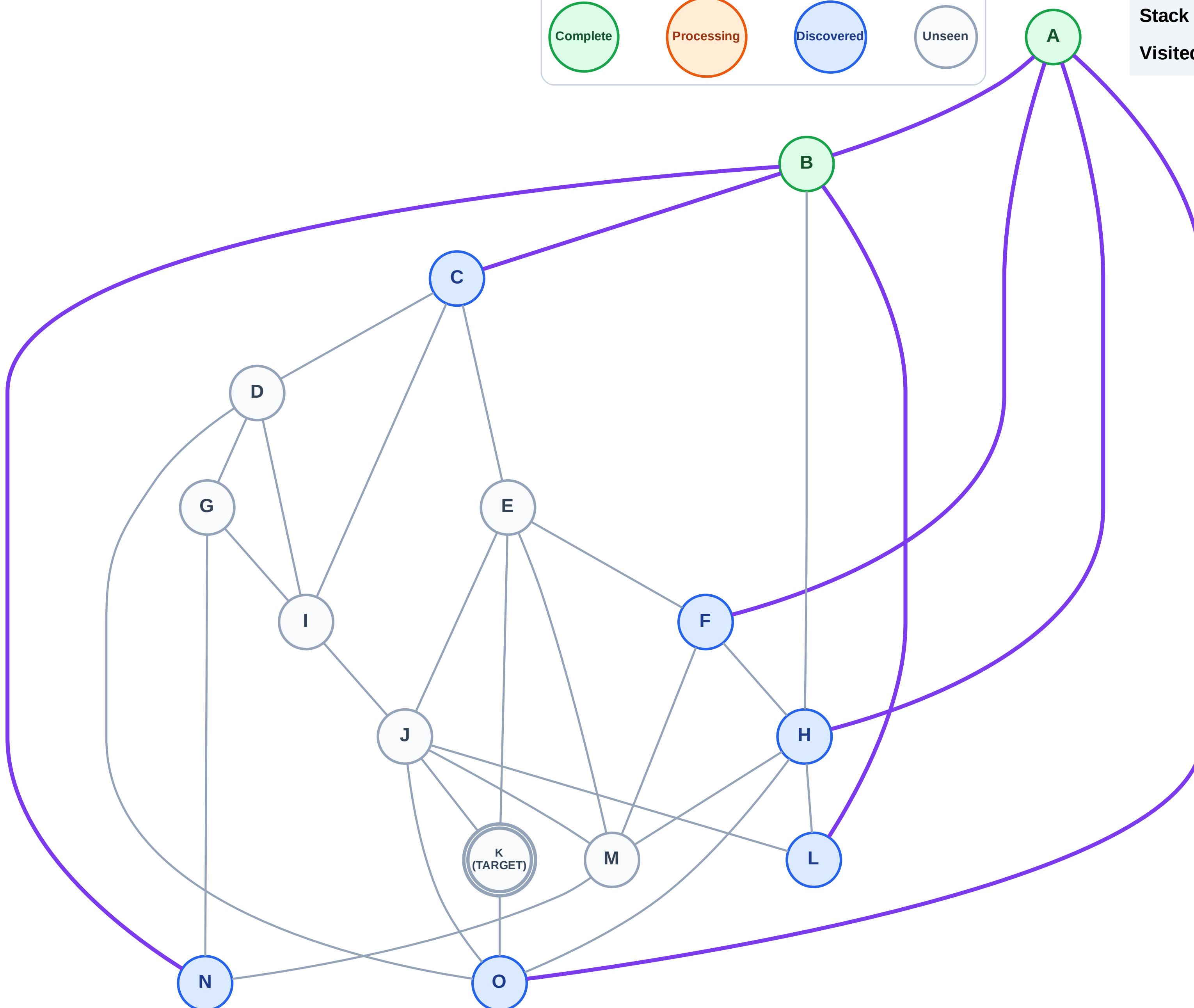
Step 5: Discover N, L, C from B

Legend



Stack (top at right): O | H | F | N | L | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

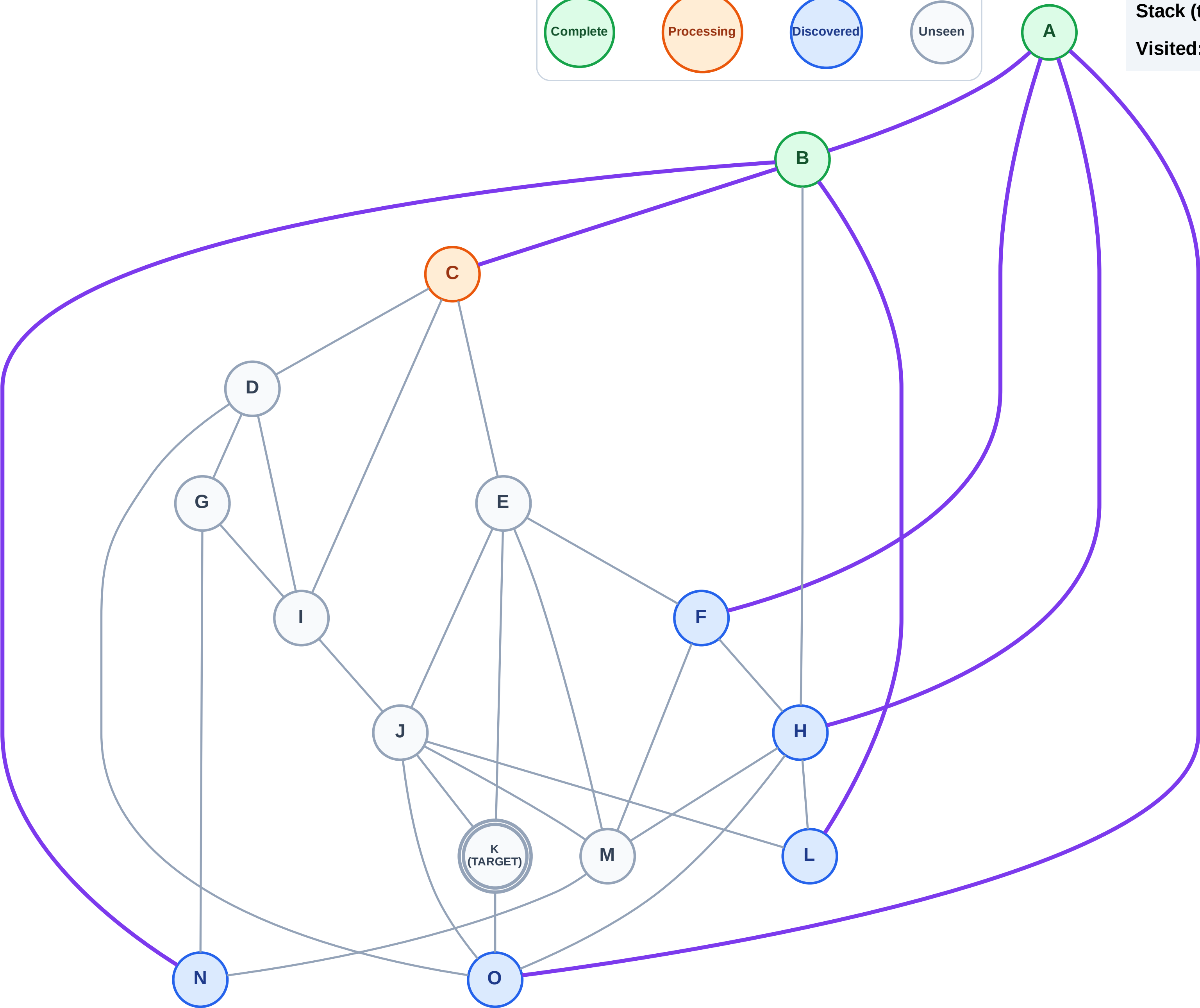
Complete

Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L
Visited: A, B

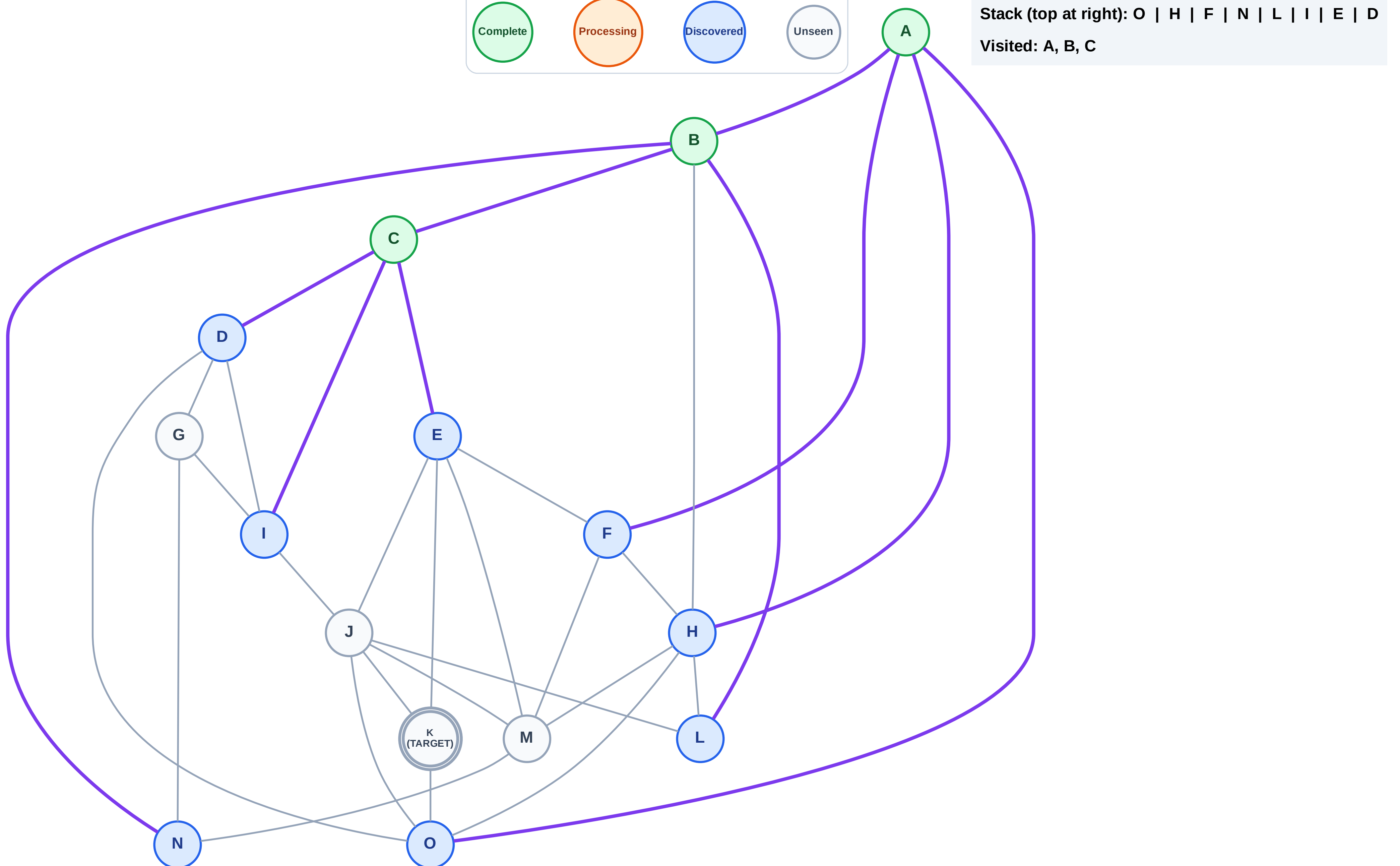


Step 7: Discover I, E, D from C

Legend

Unseen

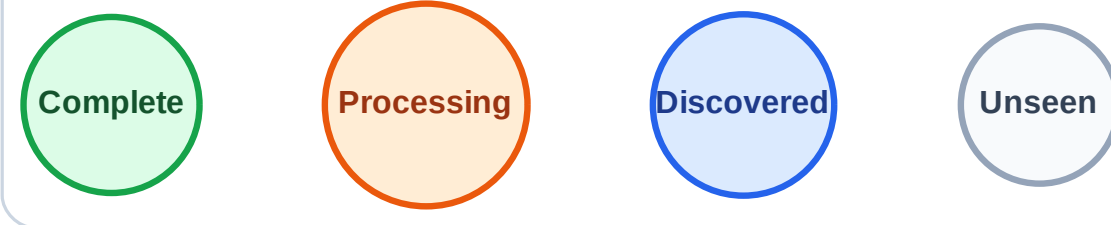
Visited: A, B, C



DFS Traversal

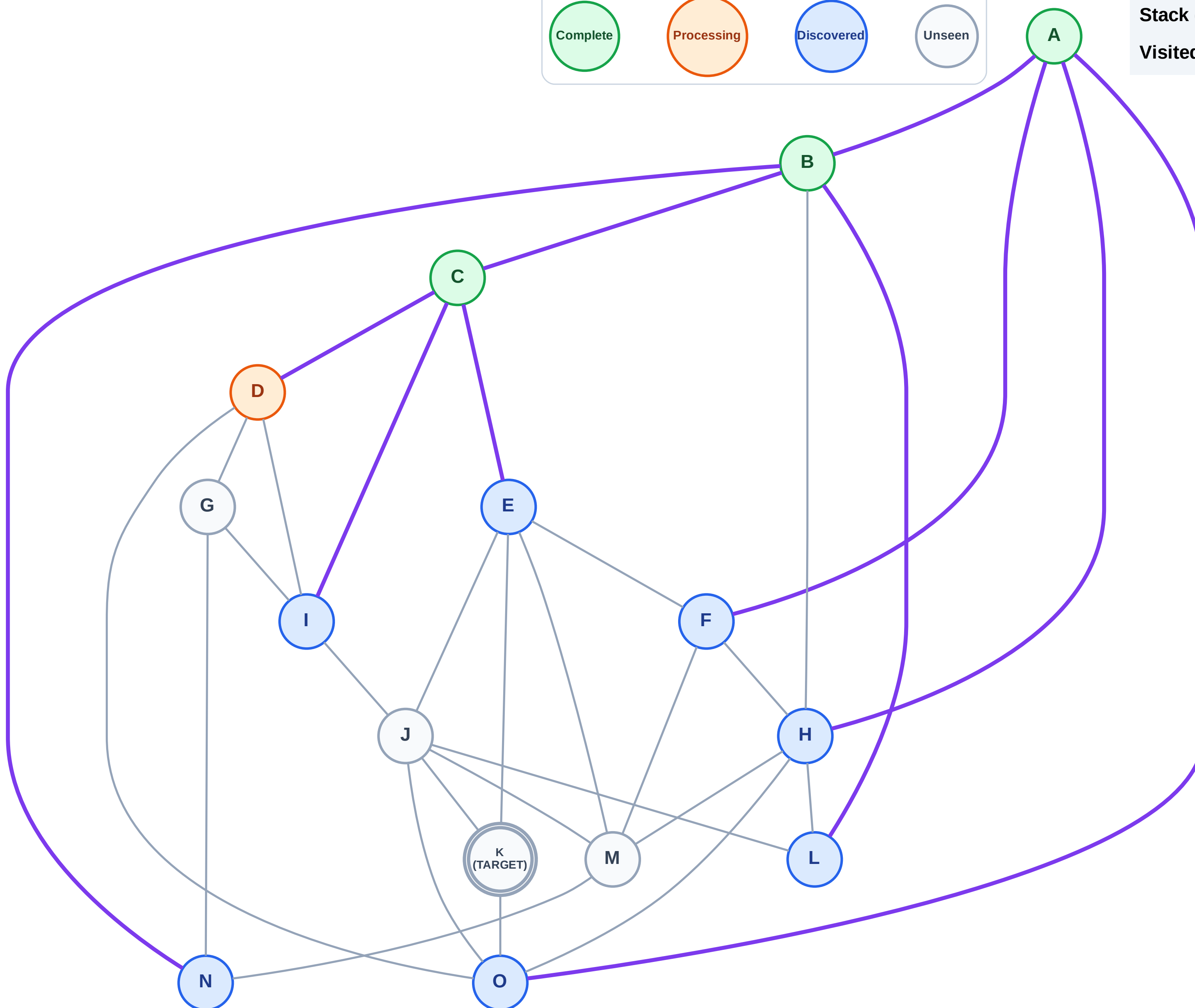
Step 8: Process node D

Legend



Stack (top at right): O | H | F | N | L | I | E

Visited: A, B, C



DFS Traversal

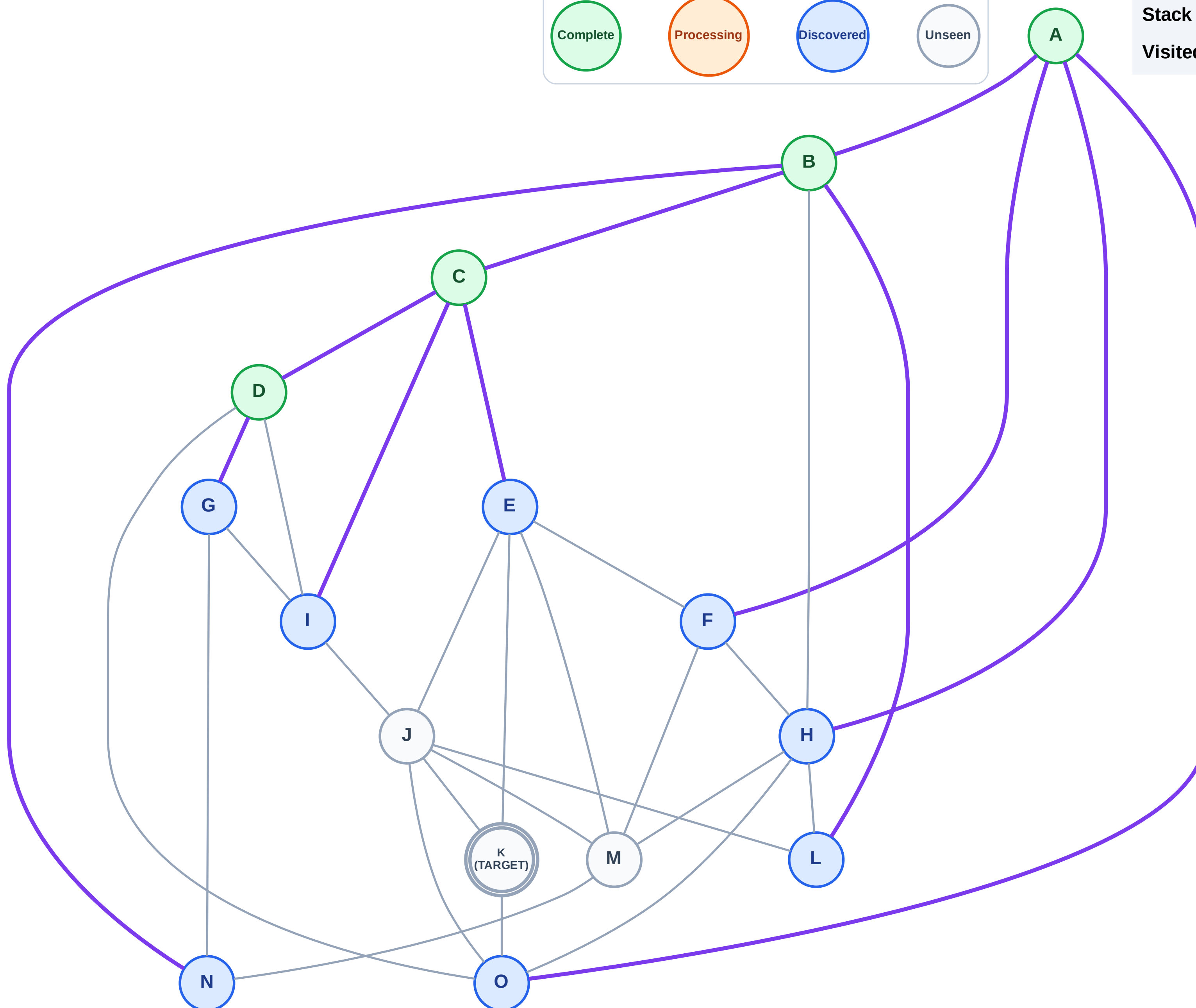
Step 9: Discover G from D

Legend



Stack (top at right): O | H | F | N | L | I | E | G

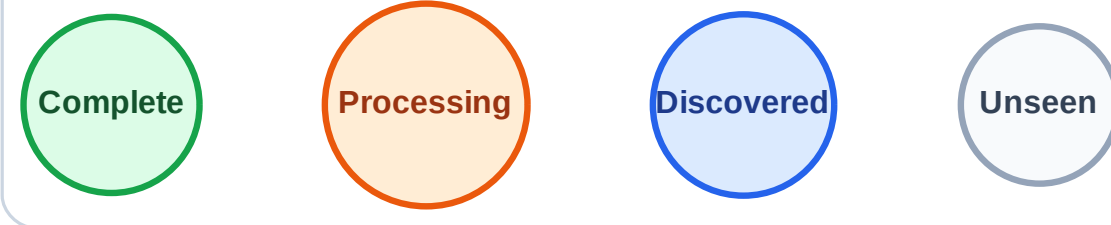
Visited: A, B, C, D



DFS Traversal

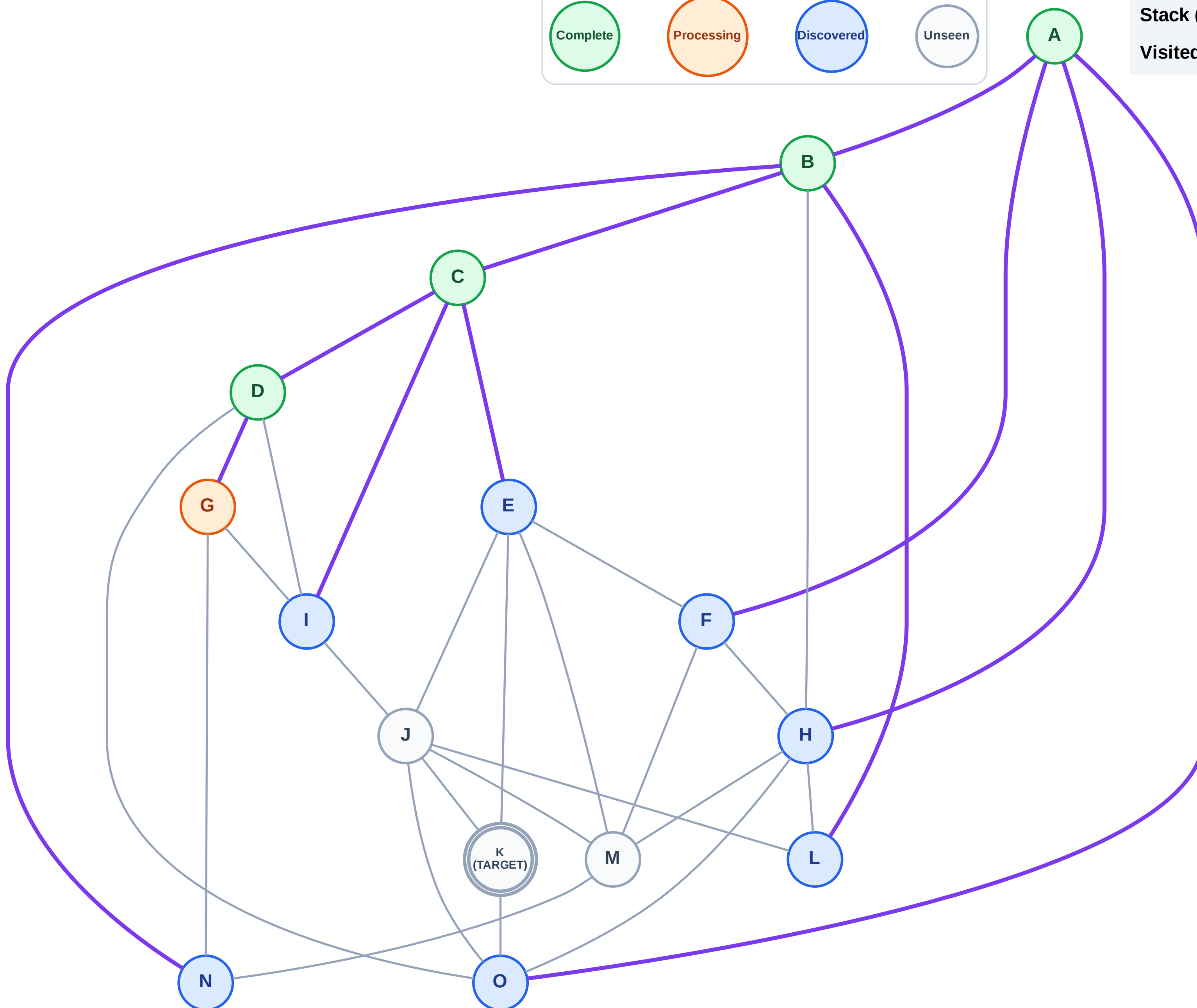
Step 10: Process node G

Legend



Stack (top at right): O | H | F | N | L | I | E

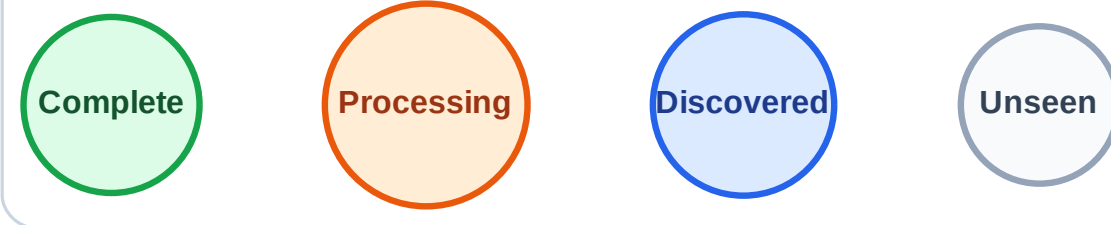
Visited: A, B, C, D



DFS Traversal

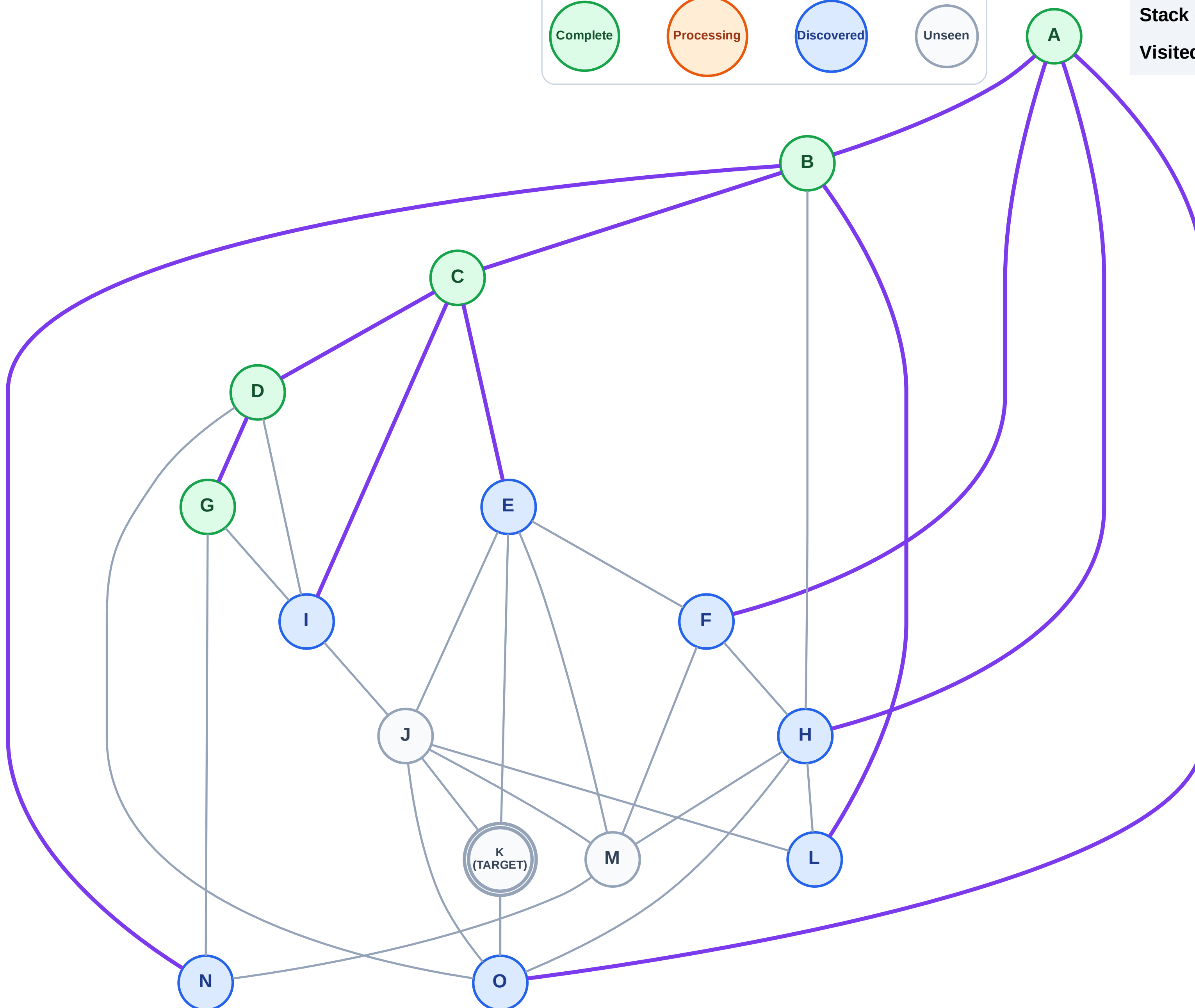
Step 11: No new neighbors from G

Legend



Stack (top at right): O | H | F | N | L | I | E

Visited: A, B, C, D, G



Step 12: Process node E

Step 12: Process node E

Legend

Complete

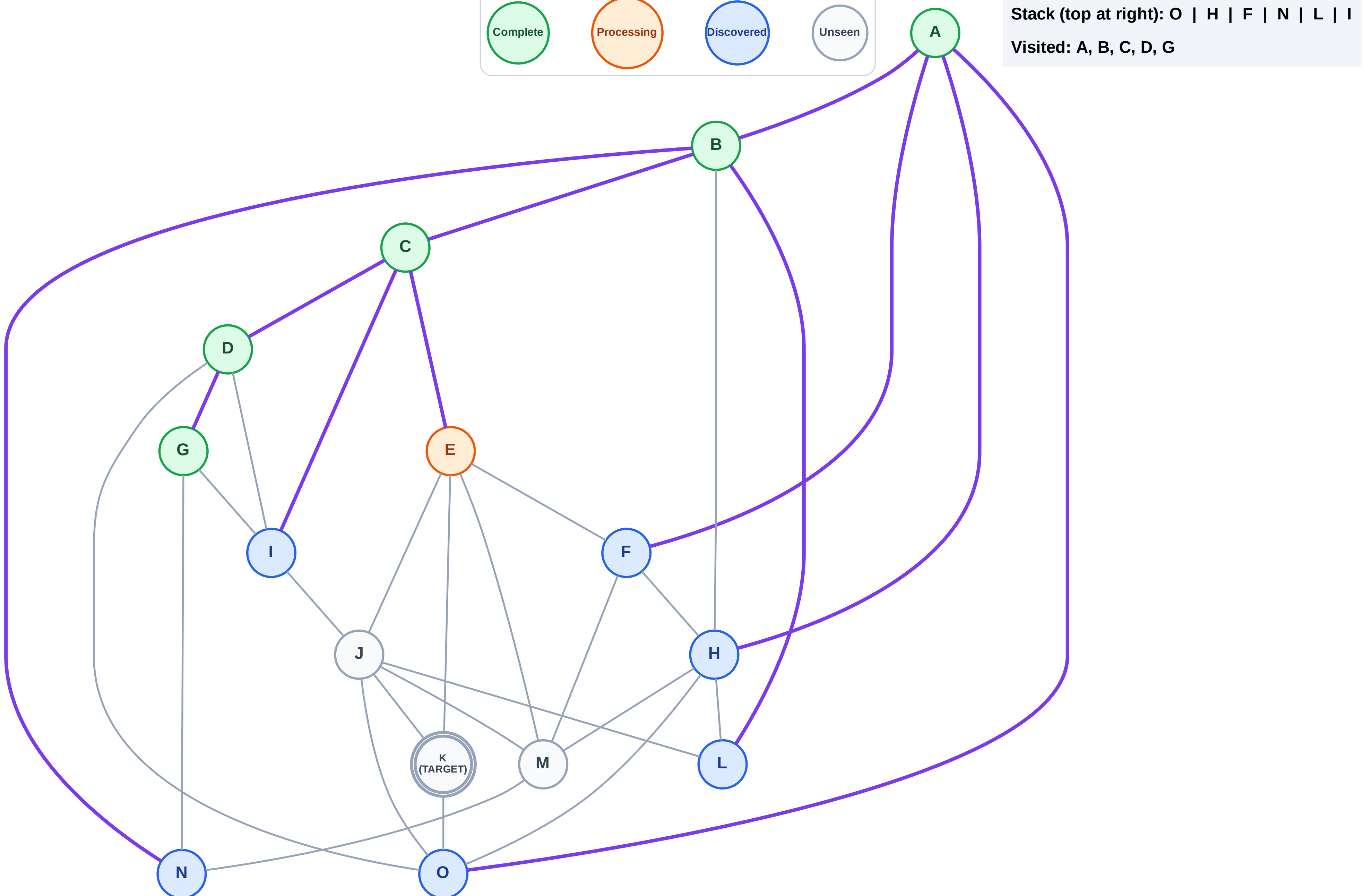
Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I

Visited: A, B, C, D, G



DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I | M

Visited: A, B, C, D, E, G, K

